

# NEST

CSE495 — IT Project Management & Entrepreneurship

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# Problem Identification

## **A short-visit scenario**

A visitor arrives in Dhaka several hours before a job interview, appointment, examination or other activity and needs a place to rest, freshen up or prepare.

## **Current challenge**

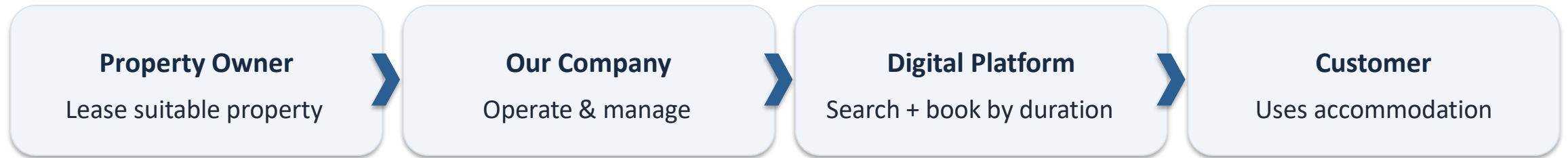
Conventional accommodation is generally structured around longer stays. Short-duration/day-use options do exist, but they are not uniformly presented as a standard booking product.

## **Opportunity**

Can a dedicated accommodation operator make flexible-duration booking a core service, with clear pricing, digital availability and controlled room turnover?

# Proposed Solution

A company-operated accommodation model supported by a dedicated booking platform.



## Private Rooms

Flexible-duration room booking with availability and automatic price calculation.

## Dormitories

Short-duration shared accommodation with a proposed minimum booking of 2 hours.

# Target Customers

## **Job applicants**

Interview preparation

## **Students**

Admission / exams

## **Medical visitors**

Appointments / treatment visits

## **Business travelers**

Meetings / work

## **Transit travelers**

Layovers / short gaps

## **Tourists & budget users**

Short temporary stay

# Market Opportunity & Competitor Context

The market is not empty; the opportunity is in how the service is organized.

## Conventional booking market

- Hotels mainly offer daily/overnight bookings.
- Customers pay for a full day, even for short stays.

## Short-stay evidence

- Hotel Fountain in Nikunja explicitly offers day-use and short transit stays.
- Day-use and transit hotels already serve customers needing accommodation for a few hours

## Our distinction

- 2-hour minimum booking
  - Affordable shared accommodation
- Pay only for the time needed
  - Easy online booking

# Business Case

## Problem

Some visitors may need accommodation for only part of a day; existing short-stay offerings are fragmented.

## Proposed Solution

Lease suitable properties, operate accommodation, and let customers book rooms/dorms by duration.

## Expected Value

Convenient duration-based booking, clearer pricing, potential capacity utilization, and a differentiated business concept.

## Feasibility Focus

Demand, lease cost, utilization, turnover cost, staffing, regulation, technical feasibility and revenue model.

## Risk

**Low utilization • High lease costs • Room turnover delays  
• Double booking • Pricing uncertainty • Regulatory requirements**

# Project Objectives

- Analyze the identified short-duration accommodation problem.
- Develop a feasible flexible-duration accommodation business concept.
- Design the core digital booking process.
- Develop a working prototype.
- Address availability, cleaning, inspection and booking-buffer requirements.
- Evaluate technical, operational and financial feasibility.

# Project Charter

| Charter Element  | Current Definition                                                              |
|------------------|---------------------------------------------------------------------------------|
| Project title    | NEST                                                                            |
| Project manager  | Belal Hossain                                                                   |
| Sponsor          | NEST                                                                            |
| Background       | Short-duration accommodation need in Dhaka (expanding to different areas later) |
| Purpose          | Evaluate feasibility and demonstrate the concept through a working prototype.   |
| Initial market   | Dhaka                                                                           |
| Operating model  | Lease suitable property → operate accommodation                                 |
| Core deliverable | Working booking prototype + project/business documentation                      |

# Stakeholders

| Stakeholder            | Interest / Concern                         | Influence |
|------------------------|--------------------------------------------|-----------|
| Supervisor             | Academic requirements, progress, quality   | High      |
| Project Team           | Planning, development, documentation       | High      |
| Customers              | Duration, price, availability, convenience | Medium    |
| Property Owners        | Lease terms, property use, reliability     | High      |
| Accommodation Staff    | Cleaning, room status, check-in/out        | Medium    |
| Management             | Profitability, service quality, growth     | High      |
| Regulatory Authorities | Applicable registration/licensing          | High      |

# High-Level Scope

## Customer Module

- Registration/login
- Search & location/date
- Room/dorm selection
- Duration selection
- Availability
- Price calculation
- Booking
- Confirmation
- Booking history
- Basic cancellation
- Customer review

## Operation & Admin

- Room management
- Availability
- Booking management
- Room status
- Check-in/out information
- Pricing management
- Basic reports
- User/property management

## Out of Scope

- Hotel construction
- Nationwide deployment
- Enterprise infrastructure
- Unnecessary advanced AI

# Major Deliverables

## Management

Business case, charter, stakeholders, RACI, scope, WBS

## Planning

Schedule, cost estimate, risk plan, quality plan

## Business

Customer segments, value proposition, competitor context, business model

## System

Requirements, user flow, architecture/database/UI design

## Prototype

Customer booking + hotel/admin management

## Validation

Functional tests, booking/pricing/availability tests

## Final Output

Report, presentation, prototype demonstration

# Initial Timeline



# Initial Risks

The most important risk is operational: flexible bookings increase room turnover.

## Demand

Low bookings / utilization

Response: Validate assumptions;  
start small

## Lease cost

Fixed cost may be too high

Response: Compare lease structures

## Turnover

Cleaning/inspection causes gaps

Response: Use operational buffers

## Pricing

Wrong duration calculation

Response: Automated rules + testing

## Regulation

Registration/licensing requirements

Response: Include compliance  
research

## Scope

Too many features

Response: Prioritize core flow

## Time

Semester constraints

Response: Parallelize work; track  
milestones

## Core booking rule

Previous booking end + operational buffer  $\leq$  next booking start

# Expected Benefits

## Customer

- Flexible duration
- Convenient digital booking
- Clear duration-based price
- Better alignment between use and booked time

## Business

- Differentiated service concept
- Potential capacity utilization
- Direct operational control
- Digital booking and management
- Potential expansion to other locations

## Academic / Project

- Business + technical integration
- Working prototype
- Feasibility analysis

# Conclusion

## What we are proposing

A Dhaka-focused accommodation business with other areas that leases suitable properties, operates rooms/dormitories, and lets customers book according to the duration they need.

## What makes it different

Flexible-duration accommodation is the core product, supported by digital availability, pricing and booking logic — not merely an occasional day-use package.

## Confirmed in the Project

Dhaka • leased/operated properties • private rooms + dorms • flexible duration • current pricing assumptions

## Project

NEST • PM- Belal Hossain • Lease a building and manage it for our hotels\dormitories • pilot location- Dhaka

THANK YOU